

B.Sc. Engg. SWE 5th Semester

03 October 2022 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2021-2022
FULL MARKS: 75

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper
Answer all 3 (three) questions. Marks of each question and corresponding CO and PO are written in the right margin.

1. a) Suppose you have a spherical oil tank that is 6 ft in diameter. After using up some oil from your tank, you want to know how much oil is left in the tank. One way of getting this information is to design such a ruler that you can dip inside the tank and based on the height of the ruler that becomes wet with oil, you can find the volume of oil left in the tank.

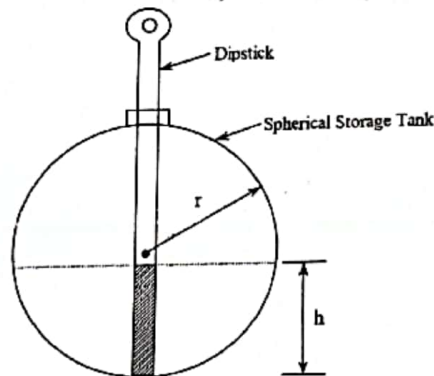


Figure 1: Oil in a spherical storage tank

Suppose you found that the relationship between the volume of oil and the height of the ruler that will be marked with oil is given as:

$$V = \frac{\pi h^2 (3r - h)}{3}$$

where r is the radius of the spherical tank.

For marking the different heights of the ruler, you would have to solve the above nonlinear equation for different volumes. Suppose you wanted to use a bracketing method like Regula Falsi, otherwise known as the False Position Method.

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| i. Formulate the given scenario as a nonlinear equation problem that can be solved using the False Position Method. | 4
(CO1)
(PO2) |
| ii. Apply the False Position method and perform at least 4 iterations to find the height at which you would mark the scale for measuring 4 ft ³ of oil. | 10
(CO2)
(PO1) |
| iii. Analyze your results to see if you are converging towards the solution by checking the absolute relative approximate error at each iteration. | 3
(CO3)
(PO2) |
| b) Differentiate between Round Off Error and Truncation Error? What are the different ways Truncation error may be introduced? | 5+3
(CO3)
(PO2) |

2. a) i. The Maclaurin series is a special case of Taylor series. Find the Maclaurin series for $\cos(x)$ starting from the general definition of Taylor Series. (CO1) (PO2)

- ii. Estimate the bounds of the truncation error in the representation of $\cos 2$ using the Remainder Theorem, given that only the first 3 terms of the Maclaurin series are used. (CO3) (PO2)

- b) The upward velocity of a rocket is given as a function of time in Table 1:
Table 1: Velocity as function of time for Question 2.(b)

T (s)	V (ms ⁻¹)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

Solve the following problems using the spline interpolation method with linear splines:

- i. What is the velocity of the rocket at $t = 14$? (CO2) (PO1)

- ii. What is the acceleration of the rocket at $t = 21$? (CO2) (PO1)

- iii. What is the distance travelled by the rocket from $t = 14$ to $t = 21$? (CO2) (PO1)

3. a) Find the Quadratic form of Newton's Divided Difference polynomial and using its pattern, find the Cubic form of Newton's Divided Difference polynomial. (CO1) (PO2)

- b) Compute $f(6)$, $f'(6)$ and $f''(6)$ given that $f(5) = -30$, $f'(5) = 10$, $f''(5) = 16$ and $f'''(5) = 6$ and all higher derivatives of $f(x)$ at $x = 5$ are zero. (CO2) (PO1)